Disentangled Language Representation:
From Motif Vectors to Causal Process IR

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Abstract

This paper reports a sequence of eight experiments on Disentangled Language Representation (DLR). The central claim is not that all language can be compressed into 32 dimensions. Rather, natural language can be decomposed into multiple layers: a low-dimensional structural motif basis, typed role and transition frames, causal frame graphs, and process-level trace diagnoses with latent preconditions. Early experiments show that a 32-dimensional motif vector can group paraphrases, idioms, and cross-lingual expressions more cleanly than a surface lexical baseline. Later experiments show why motif vectors alone are insufficient: they miss role inversion, authority legitimacy, temporal status, hidden invariants, and local-versus-global success. The final experiments move from sentence similarity to trace diagnosis. Synthetic multi-event traces are compiled into ProcessIR objects that identify false terminal states, violated invariants, authority failures, reconciliation failures, decay sources, and latent structural preconditions. Across the implemented benchmark suite, DLR progresses from semantic similarity scoring to a typed causal intermediate representation for structural language understanding.

1 Introduction

Modern embedding systems represent utterances as high-dimensional vectors that mix structure, parameters, bindings, and expression into one space. This is useful for semantic retrieval, but it makes a different question difficult: what structural process does an utterance describe?

For example, *He was fired from his job*, *He was given the axe*, and *Le han dado la puerta* differ in surface form, idiom, and language, yet they describe the same structural event: an authority removes an agent from an institutional role boundary. Conversely, *The bank approved my loan* and *The bank rejected my loan* share most words and entities, but differ in outcome polarity inside the same permission-gate schema.

The DLR project tests whether language can be decomposed into:

1. a 32-dimensional structural motif basis;
2. scalar parameters such as intensity, urgency, polarity, and confidence;
3. entity and role bindings;
4. typed transitions, authority, boundary, invariant, and process status;
5. causal graphs and process traces.

The experiments began as sentence-pair similarity tests and evolved into process diagnosis benchmarks. This paper presents the implementation and results of DLR-1 through DLR-8.

2 Motif Basis

The structural vocabulary uses 32 universal motifs grouped into six families. These motifs are not intended to encode every fact about language. They provide a basis for structural skeletons, while parameters, bindings, expressions, frames, and graphs carry the rest of the meaning.

Table 1: The 32 motif basis.

Group	Motifs
Existence	State, Transition, Invariant, Identity, Boundary, Terminal State, Decay
Persistence	Storage, Addressing, Replication, Synchronization
Intelligence	Representation, Feedback, Prediction, Search, Model, Compression, Optimization, Explore/Exploit, Self-Reference
Structure	Composition, Hierarchy, Modularity, Abstraction, Emergence
Allocation	Scarcity, Queue, Scheduling
Coordination	Communication, Authority, Reconciliation, Negotiation

3 Representation Evolution

DLR intentionally became more structured over time. Table 2 summarizes the progression.

Table 2: Evolution of the DLR representation.

Stage	Object	New capability
DLR-1	Motif vector	Tests whether paraphrases and idioms share a structural skeleton.
DLR-2	Typed frame score	Separates same schema from opposite outcome and detects polysemy.
DLR-3	StructuralIR	Adds role slots, transition family, polarity, constraints, and expected relation labels.
DLR-4	Robust StructuralIR	Adds confidence, ambiguity, candidate frames, negation, modality, and pragmatics.
DLR-5	Adversarial IR	Adds parse modes, authority legitimacy, event status, invariant status, process scope, and composite frame graphs.
DLR-6	Causal Frame Graph	Adds typed causal node roles and typed causal edges.
DLR-7	ProcessIR	Diagnoses timestamped multi-event traces with false terminal states and root causes.
DLR-8	ProcessIR plus latent conditions	Adds hidden preconditions, latent nodes, and boundary/identity/decay diagnosis.

4 Experimental Suite

The experiments are synthetic and deterministic. They are designed as falsification probes, not as claims of broad corpus performance. The purpose is to test whether the representation has the

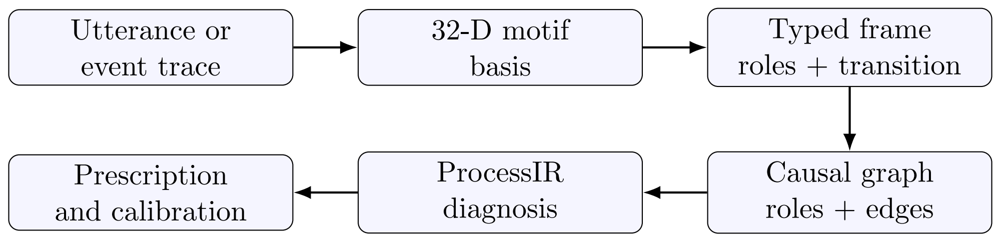

Figure 1: DLR moves from surface utterances to typed process diagnosis.

right failure modes.

Table 3: Benchmark categories.

Category	Purpose
A-B	Same structural frame across expression, idiom, and language.
C-F	Opposite outcomes, polysemy, gate schemas, and polarity-aware frames.
G-H	Role inversion and invalid authority.
I-M	Active/passive, ambiguity, context dependence, negation, modality, sarcasm.
N-S	Adversarial IR falsification: composite metaphors, deceptive authority, invariant violations, temporal ambiguity, local/global success.
T-X	Typed causal graphs for authority, invariant, completion, and terminal outcome.
Y1-Y5	Timestamped process traces with false terminal states and root-cause diagnosis.
Z1-Z5	Hidden preconditions: boundary decay, stale representations, identity aliasing, hidden queues, stale replicas.

5 Scoring

The early DLR experiments use cosine similarity over motif vectors. Later experiments add frame and graph channels:

- motif similarity;
- transition family and transition type compatibility;
- role assignment, authority, and boundary compatibility;
- outcome and state delta compatibility;
- causal node recall and causal edge recall;
- root-cause recall;
- false-terminal and latent-condition detection.

For DLR-3 and later, polarity contributes only inside compatible transition families. This avoids giving high scores to examples such as employment firing and weapon firing merely because both include a forceful transition.

6 Results

6.1 DLR-1 and DLR-2: Motif and Typed Frame Similarity

The first experiments validated the basic distinction between lexical surface similarity and structural similarity. In categories A and B, motif vectors clustered paraphrases and idioms. In category C, typed polarity reduced the score for approval versus rejection despite shared words.

Table 4: Early similarity results.

Category	Baseline	Motif	Typed frame	Interpretation
A	0.0297	0.9988	0.9447	Hunger variants share a frame.
B	0.1024	1.0000	1.0000	Job termination idioms share a frame.
C	0.6316	0.5386	0.3434	Loan approval and rejection split by outcome.
D	0.5161	0.4528	0.2807	Hunger polysemy does not collapse blindly.
E	0.3403	0.5325	0.2768	*Fired* polysemy is separated by frame type.
F	0.3230	0.6678	0.5263	Gate schemas align while polarity remains explicit.

6.2 DLR-3 through DLR-8

Table 5: Summary of later experiment results.

Stage	Cases	Passed	Accuracy	Confidence	Main finding
DLR-3	51	51	1.0000	–	StructuralIR handles roles, authority, and polarity.
DLR-4	13	13	1.0000	0.7615	Ambiguity and modality can be calibrated.
DLR-5	13	12	0.9231	0.6746	Adversarial IR reveals one useful authority/invariant confusion.
DLR-6	10	10	1.0000	0.7950	Typed causal graphs separate five causal patterns.
DLR-7	5	4	0.8000	0.8200	ProcessIR detects all false terminal states; one high-confidence decay miss.
DLR-8	5	5	1.0000	0.8080	Latent hidden preconditions are recovered in all Z traces.

6.3 False Terminals and Latent Preconditions

The process-level experiments are the strongest evidence for why single-frame sentence similarity is insufficient. DLR-7 and DLR-8 show that a local subsystem can report success while the parent process has not reached a valid terminal state. DLR-8 further shows that the root cause may be latent: not an event in the trace, but a hidden boundary, identity, representation, queue, or replication condition inferred from the trace.

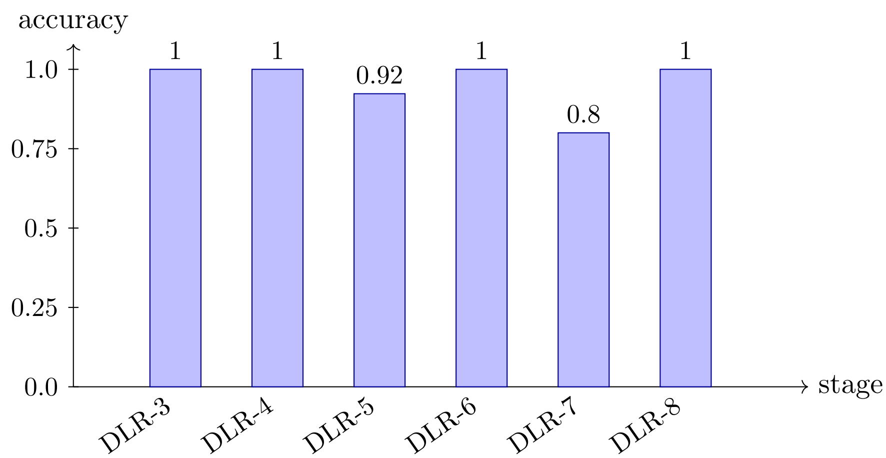

Figure 2: Stage accuracy from StructuralIR validation through hidden precondition diagnosis.

Table 6: Process-level diagnostic results.

Stage	Node recall	Edge recall	Root cause	False terminal	Overconf. fail
DLR-7	0.9667	0.9500	0.9000	1.0000	0.2000
DLR-8	1.0000	1.0000	1.0000	1.0000	0.0000

7 Examples

7.1 Role and Authority

DLR-3 adds role inversion tests such as:

> The doctor cleared him for surgery.
> He cleared the doctor for surgery.

The motif bag and transition family are similar, but the authority and role assignment are invalid. The typed IR assigns low role similarity and low final structural score.

7.2 Local Success, Global Failure

In DLR-7, the release incident trace contains:

> CI passed unit tests. CD marked deployment successful. The release bot closed the ticket. Health checks then reported a 500 spike.

The local deploy step is a transition, but the release is not terminal-valid. The ProcessIR marks the deploy success as a false terminal state, ticket closure as a premature transition, and health failure as a terminal outcome.

7.3 Latent Boundary Decay

In DLR-8, a cleanup tool deletes `/tmp/cache`. Later the workspace cache is missing. The trace does not directly contain the symlink. The gold diagnosis contains a latent condition:

Table 7: DLR-8 hidden precondition categories.

Cat.	Hidden condition	Latent kind	Motif
Z1	Symlink/path alias crosses a declared filesystem boundary.	boundary decay	Boundary, Decay
Z2	Worker uses a stale cached policy snapshot.	stale representation	Representation
Z3	Email canonicalization collapses two identities.	identity alias	Identity
Z4	Schedule success hides an execution backlog.	hidden queue	Queue, Scheduling
Z5	Read replica diverges from primary inventory state.	replication divergence	Replication, Synchronization

> `lc_path_alias`: Symlink/path alias made `/tmp/cache` resolve into a workspace-owned cache.

This is the kind of hidden precondition that a pure event log or sentence embedding tends to miss.

8 Discussion

The experiments support a narrower and stronger version of the initial hypothesis:

> The structural skeleton of many utterances can be represented with a compact motif basis, but robust reasoning requires typed frames, causal graphs, process state, and latent conditions.

The 32-dimensional motif vector is useful as a basis, not as a complete representation. It functions like a low-dimensional physics vocabulary. The complete representation is closer to an intermediate representation or abstract syntax tree for events and processes.

9 Limitations

The current benchmark is synthetic and deterministic. The parser outputs in this repository are fixtures, not learned model predictions. Therefore these experiments validate the representational target and scoring criteria, not yet an independently trained parser. The next step is to replace hand-authored IR fixtures with model-generated parses and evaluate whether the same scoring suite exposes reliable calibration, ambiguity handling, and causal diagnosis.

The reported perfect scores in some stages should not be read as generalization claims. They indicate that the implemented IR is expressive enough for the current targeted cases. The useful failures, such as the DLR-5 deceptive authority versus invariant violation confusion and the DLR-7 agentic decay miss, are more informative than raw accuracy.

10 Reproducibility

The repository includes executable TypeScript experiments and generated JSON artifacts. The main commands are:

```
npm install
npm run typecheck
npm run run:report
npm run run:dlr3
npm run run:dlr4
npm run run:dlr5
npm run run:dlr6
npm run run:dlr7
npm run run:dlr8
```

The generated reports are:

- `dlr_results.md`
- `dlr3_structural_ir.md`
- `dlr4_parser_robustness.md`
- `dlr5_adversarial_ir_falsification.md`
- `dlr6_causal_frame_graph.md`
- `dlr7_trace_to_causal_ir.md`
- `dlr8_hidden_preconditions.md`

11 Conclusion

DLR began with a simple question: can language be factored into a compact structural skeleton plus parameters, bindings, and expression? The answer from these experiments is yes, but only if the skeleton is treated as the first layer of a richer typed IR. Motif vectors group structural paraphrases. Typed frames separate role, authority, boundary, and outcome. Causal frame graphs distinguish invalid authority, invariant violation, blocked attempts, and downstream failure. ProcessIR diagnoses false terminal states in traces. Latent conditions capture hidden boundary decay, stale representation, identity aliasing, hidden queues, and replication divergence.

The practical direction is clear: train or prompt parsers to emit calibrated StructuralIR and ProcessIR objects, then evaluate them with adversarial graph and trace benchmarks rather than only embedding similarity.